

The two lost notes on page 54 of Ramanujan's lost notebook

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Abstract

On page 54 of his lost notebook Ramanujan states two families of cubic theta-function identities and, in each family, writes “see note” next to one further claim ($A^3 - B^3$ in the first family, $A^3 + B^3$ in the second). Both notes are lost: Andrews and Berndt record that they have not been preserved. We show that both missing identities are special cases of another claim written on the same page, Ramanujan's cubic circular summation (Entry 8.2.3 in Andrews–Berndt), and that in terms of the Borweins' cubic theta function $a(q) = \sum_{m,n} q^{m^2+mn+n^2}$ they take the compact form $A^3 - B^3 = q^2 f^3(-q^3, -q^{12}) + f(-q^2, -q^3) a(q^5)$ and $q(A^3 + B^3) = f^3(-q^6, -q^9) - f(-q, -q^4) a(q^5)$. We then prove why a note was unavoidable: unlike the four cube identities Ramanujan recorded, the two deferred combinations vanish in the upper half-plane. The second one vanishes at a real point $q_0 \in (-\frac{7}{10}, -\frac{1}{2})$ (an intermediate-value argument with exact rational bounds), and the conjugation $u_2(\gamma\tau)u_1(\tau) = -1$ by $\gamma = (\frac{7}{15} \frac{6}{13}) \in \Gamma_0(15)$, proved with Siegel functions, carries this zero to a zero of $A - B$ in the first family. Since a product of theta functions never vanishes, neither deferred combination—in contrast to every one-term identity on the page—is a theta product in any basis. A finite search shows that, within the basis of page 54, the only two-term product forms are the trivial rewritings $A^3 \mp B^3 = (A^3 \pm B^3) \mp 2B^3$ of the recorded one-term identities. All identities are certified to **2000** coefficients by exact integer arithmetic and independently at 50-digit precision.

Keywords: Ramanujan's lost notebook, theta functions, cubic theta functions, circular summation, Borwein cubic theta function, Siegel functions

MSC Classification: 11F27 , 33E05 , 11F20 , 11F03 , 05A30

1 Introduction

As usual, let

$$f(a, b) := \sum_{n=-\infty}^{\infty} a^{n(n+1)/2} b^{n(n-1)/2}, \quad |ab| < 1,$$

and $f(-q) := f(-q, -q^2) = (q; q)_{\infty}$. Page 54 of Ramanujan's lost notebook [1] contains two groups of cubic identities for theta functions with quotient of arguments q^{15} ; they are treated as Entries 8.2.1 and 8.2.2 in Andrews and Berndt's second volume [2, pp. 176–178].

Entry 8.2.1. Let $A = f(-q^7, -q^8)$ and $B = qf(-q^2, -q^{13})$. Then

$$A + B = \frac{f(-q^2, -q^3)}{f(-q, -q^4)} f(-q^5), \quad (8.2.6)$$

$$A - B = f(-q^{2/3}, -q) + q^{2/3} f(-q^3, -q^{12}), \quad (8.2.7)$$

$$A^3 + B^3 = \frac{f(-q^6, -q^9)}{f(-q^3, -q^{12})} f^3(-q^5), \quad (8.2.8)$$

$$(A - B)^3 = \frac{f(-q^2, -q^3) f^3(-q, -q^4)}{f(-q^3, -q^{12})} + q^2 f^3(-q^3, -q^{12}). \quad (8.2.9)$$

After stating these, Ramanujan writes “ $A^3 - B^3$ see note.” Andrews and Berndt comment: “Either Ramanujan never wrote this note, or if he did, it has not been preserved” [2, p. 177].

Entry 8.2.2. Let $A = f(-q^4, -q^{11})$ and $B = qf(-q, -q^{14})$. Then

$$A - B = \frac{f(-q, -q^4)}{f(-q^2, -q^3)} f(-q^5), \quad (8.2.10)$$

$$A + B = -\frac{1}{q^{1/3}} \left\{ f(-q^{1/3}, -q^{4/3}) - f(-q^6, -q^9) \right\}, \quad (8.2.11)$$

$$A^3 - B^3 = \frac{f(-q^3, -q^{12})}{f(-q^6, -q^9)} f^3(-q^5), \quad (8.2.12)$$

$$(A + B)^3 = -\frac{1}{q} \left\{ \frac{f(-q, -q^4) f^3(-q^2, -q^3)}{f(-q^6, -q^9)} - f^3(-q^6, -q^9) \right\}. \quad (8.2.13)$$

Here Ramanujan writes “ $A^3 + B^3$ see note,” and again “he apparently did not write this note, or, if he did, it has been lost” [2, p. 178].

To our knowledge the two missing notes have not been reconstructed in the literature. The cubic identities of page 54 were first proved by Son [12], who already remarks that Ramanujan “apparently did not write these notes” [12, §4]; Andrews and Berndt state that in this section they “follow precisely Son's beautiful proofs” [2, p. 173], and both “see note” remarks, with the statements that the notes are lost, appear inside that very section. We were unable to find any later source addressing the notes.

This paper has two parts. The first (Section 2) observes that the two missing identities are hidden two lines further down the same page: they are special cases of

Ramanujan's cubic circular summation, Entry 8.2.3 of [2], whose right-hand side is the Borweins' cubic theta function $a(q)$; Section 3 records a second, elementary form obtained from (8.2.7), (8.2.9), (8.2.11), (8.2.13). The second part (Section 4) explains why Ramanujan could not record these two claims in the one-line product form of (8.2.8) and (8.2.12): we prove that $A^3 + B^3$ of Entry 8.2.2 vanishes at a real point $q_0 \in (-\frac{7}{10}, -\frac{1}{2})$, that an explicit element of $\Gamma_0(15)$ conjugates the two families and carries this zero to a zero of $A - B$ in Entry 8.2.1, and that consequently neither deferred combination is a product of theta functions in any basis (Theorem 8 and Corollary 9). Section 5 adds a finite search showing that within the basis of page 54 the only two-term product forms are trivial rewritings of the recorded identities, and Section 6 describes the verification scripts.

2 The notes as circular summations

On page 54 Ramanujan also states the following, proved by Son [4] and given as Entry 8.2.3 in [2, p. 178]: for $|ab| < 1$ and $q = ab$,

$$f^3(ab^2, a^2b) + af^3(b, a^3b^2) + bf^3(a, a^2b^3) = f(a, b) \left\{ \frac{f^9(-q)}{f^3(-q^3)} + 27q \frac{f^9(-q^3)}{f^3(-q)} \right\}^{1/3}. \quad (1)$$

With the Borweins' cubic theta functions [7, 8]

$$a(q) = \sum_{m,n=-\infty}^{\infty} q^{m^2+mn+n^2}, \quad b(q) = \frac{f^3(-q)}{f(-q^3)}, \quad c(q) = 3q^{1/3} \frac{f^3(-q^3)}{f(-q)},$$

which satisfy $a^3 = b^3 + c^3$ [7], the brace in (1) equals $(b^3(q) + c^3(q))^{1/3} = a(q)$.

Theorem 1 Let $A = f(-q^7, -q^8)$ and $B = qf(-q^2, -q^{13})$ (Entry 8.2.1). Then

$$A^3 - B^3 = q^2 f^3(-q^3, -q^{12}) + f(-q^2, -q^3) a(q^5). \quad (2)$$

Proof Take $a = -q^3$, $b = -q^2$ in (1), so $ab = q^5$. Then $f(ab^2, a^2b) = f(-q^7, -q^8) = A$, $af^3(b, a^3b^2) = -q^3 f^3(-q^2, -q^{13}) = -B^3$, $bf^3(a, a^2b^3) = -q^2 f^3(-q^3, -q^{12})$ and $f(a, b) = f(-q^2, -q^3)$. $\square$

Theorem 2 Let $A = f(-q^4, -q^{11})$ and $B = qf(-q, -q^{14})$ (Entry 8.2.2). Then

$$A^3 + B^3 = \frac{1}{q} \left\{ f^3(-q^6, -q^9) - f(-q, -q^4) a(q^5) \right\}. \quad (3)$$

Proof Take $a = -q^6$, $b = -q^{-1}$ in (1), so again $ab = q^5$. Then $f(ab^2, a^2b) = f(-q^4, -q^{11}) = A$. By $f(a, b) = a f(a^{-1}, a^2b)$ [3, p. 34],

$$f(-q^{-1}, -q^{16}) = -q^{-1} f(-q, -q^{14}), \quad f(-q^{-1}, -q^6) = -q^{-1} f(-q, -q^4);$$

hence $af^3(b, a^3b^2) = -q^6 f^3(-q^{-1}, -q^{16}) = q^3 f^3(-q, -q^{14}) = B^3$, $bf^3(a, a^2b^3) = -q^{-1} f^3(-q^6, -q^9)$ and $f(a, b) = -q^{-1} f(-q, -q^4)$. $\square$

Theorems 1–2 have exactly the shape of the recorded identities (8.2.9) and (8.2.13) — a cube of a level-15 theta function plus one further term — except that the second term contains the binary theta series $a(q^5)$ rather than a theta product. Since (1) is printed on the same page, immediately after the two “see note” remarks, it is natural to read Theorems 1–2 as the content of the lost notes; this reading is an interpretation, and we label it as such. Section 4 shows that some non-product factor is in fact unavoidable.

Literature. We found no statement of Theorems 1–2, nor any link between Entry 8.2.3 and the two “see note” remarks, in [12], in [2], in the chapter on circular summation of [5], in [6], in Chan–Liu–Ng [9], or in Ge–Luo [10]. Son’s paper [4] was checked through its zbMATH review (Zbl 1062.11023), which lists the general theorem and the cases F_n , $n = 2, 3, 4, 5, 7$, and through the account of its proof in [5, Chapter 9].

3 An elementary second form

Theorem 3 *Let $A = f(-q^7, -q^8)$ and $B = qf(-q^2, -q^{13})$, as in Entry 8.2.1. Then*

$$A^3 - B^3 = \frac{f(-q^2, -q^3) f^3(-q, -q^4)}{f(-q^3, -q^{12})} + q^2 f^3(-q^3, -q^{12}) + 3AB \left\{ f(-q^{2/3}, -q) + q^{2/3} f(-q^3, -q^{12}) \right\}, \quad (4)$$

where, by the Jacobi triple product,

$$AB = q f(-q^7, -q^8) f(-q^2, -q^{13}) = q (q^2, q^7, q^8, q^{13}; q^{15})_{\infty} (q^{15}; q^{15})_{\infty}^2.$$

Proof By the polynomial identity $A^3 - B^3 = (A - B)^3 + 3AB(A - B)$, the claim is precisely (8.2.9) plus $3AB$ times (8.2.7), both of which are proved in [2, pp. 176–177]. The product form of AB is two applications of the Jacobi triple product $f(a, b) = (-a; ab)_{\infty} (-b; ab)_{\infty} (ab; ab)_{\infty}$. $\square$

Theorem 4 *Let $A = f(-q^4, -q^{11})$ and $B = qf(-q, -q^{14})$, as in Entry 8.2.2. Then*

$$A^3 + B^3 = -\frac{1}{q} \left\{ \frac{f(-q, -q^4) f^3(-q^2, -q^3)}{f(-q^6, -q^9)} - f^3(-q^6, -q^9) \right\} + \frac{3AB}{q^{1/3}} \left\{ f(-q^{1/3}, -q^{4/3}) - f(-q^6, -q^9) \right\}, \quad (5)$$

where

$$AB = q f(-q^4, -q^{11}) f(-q, -q^{14}) = q (q, q^4, q^{11}, q^{14}; q^{15})_{\infty} (q^{15}; q^{15})_{\infty}^2.$$

Proof $A^3 + B^3 = (A + B)^3 - 3AB(A + B)$; substitute (8.2.13) and (8.2.11), proved in [2, p. 178]. $\square$

Both right-hand sides were verified coefficient-by-coefficient to 2000 terms (Section 6). We emphasize what is, and what is not, claimed: the two theorems are elementary consequences of Ramanujan's own four recorded identities, and in that sense they are certainly results Ramanujan possessed. Comparing (4) with (2) gives, as a by-product, $3AB(A - B) = f(-q^2, -q^3)\{a(q^5) - f^3(-q, -q^4)/f(-q^3, -q^{12})\}$ for the pair of Entry 8.2.1, and in the same way $3AB(A + B) = q^{-1}f(-q, -q^4)\{a(q^5) - f^3(-q^2, -q^3)/f(-q^6, -q^9)\}$ for the pair of Entry 8.2.2.

4 Why a note was necessary: the deferred combinations vanish

Throughout, $\mathcal{H}$ is the upper half-plane, $q = e^{2\pi i\tau}$, $\tau \in \mathcal{H}$, and $q^r := e^{2\pi i r\tau}$ for rational r . To separate the two families we write $A_1 = f(-q^7, -q^8)$, $B_1 = qf(-q^2, -q^{13})$ (Entry 8.2.1) and $A_2 = f(-q^4, -q^{11})$, $B_2 = qf(-q, -q^{14})$ (Entry 8.2.2), and we put

$$u_1 := \frac{B_1}{A_1} = q \frac{(q^2, q^{13}; q^{15})_\infty}{(q^7, q^8; q^{15})_\infty}, \quad u_2 := \frac{B_2}{A_2} = q \frac{(q, q^{14}; q^{15})_\infty}{(q^4, q^{11}; q^{15})_\infty}, \quad (6)$$

the common factor $(q^{15}; q^{15})_\infty$ having cancelled by the Jacobi triple product. By a *theta product* we mean a function

$$P(\tau) = c q^s \prod_{i=1}^m f(\pm q^{a_i}, \pm q^{b_i})^{k_i}, \quad c \in \mathbb{C}^\times, \quad s \in \mathbb{Q}, \quad a_i, b_i \in \mathbb{Q}_{>0}, \quad k_i \in \mathbb{Z}, \quad (7)$$

with any choice of signs; every one-term identity on page 54 — (8.2.6), (8.2.8), (8.2.10), (8.2.12), and the products of Theorems 3–4 — has this form.

Lemma 5 *A theta product (7) has no zeros in $\mathcal{H}$. Moreover, for real $q \in (-1, 0) \cup (0, 1)$, with real roots q^r , it is nonzero, and real when c is real.*

Proof By the Jacobi triple product, $f(\pm q^a, \pm q^b) = (\mp q^a; q^{a+b})_\infty (\mp q^b; q^{a+b})_\infty (q^{a+b}; q^{a+b})_\infty$ is an absolutely convergent product of factors $1 \mp q^r$ with $r > 0$ rational, each of modulus in $(0, 2)$ since $|q^r| < 1$. For real q and real roots each factor lies in $(0, 2)$, and q^s is real. $\square$

4.1 The two families are conjugate in $\Gamma_0(15)$

We use the Siegel functions of Kubert and Lang [11, Chapter 2]: for $\mathbf{a} = (a_1, a_2) \in \mathbb{Q}^2$, $z = a_1\tau + a_2$, $q_z = e^{2\pi i z}$ and $B_2(x) = x^2 - x + \frac{1}{6}$,

$$g_{\mathbf{a}}(\tau) = -q^{B_2(a_1)/2} e^{\pi i a_2(a_1-1)} (1 - q_z) \prod_{n \geq 1} (1 - q^n q_z)(1 - q^n q_z^{-1}). \quad (8)$$

They satisfy, for $\gamma \in SL_2(\mathbb{Z})$ and $\mathbf{b} \in \mathbb{Z}^2$ [11, Chapter 2, (K1), (K2)],

$$g_{\mathbf{a}}(\gamma\tau) = \varepsilon(\gamma) g_{\mathbf{a}\gamma}(\tau), \quad g_{\mathbf{a}+\mathbf{b}}(\tau) = \varepsilon(\mathbf{a}, \mathbf{b}) g_{\mathbf{a}}(\tau), \quad g_{-\mathbf{a}} = -g_{\mathbf{a}}, \quad (9)$$

where $\varepsilon(\gamma)$ is a twelfth root of unity independent of $\mathbf{a}$ (it comes from η^2), $\mathbf{a}\gamma$ is the row-vector action, and $\varepsilon(\mathbf{a}, \mathbf{b}) = (-1)^{b_1 b_2 + b_1 + b_2} e^{-\pi i (b_1 a_2 - b_2 a_1)}$. In particular a quotient $g_{\mathbf{a}}/g_{\mathbf{b}}$ transforms under $SL_2(\mathbb{Z})$ without any root of unity.

Lemma 6 With $g_r := g_{(r/15, 0)}(15\tau)$,

$$u_1(\tau) = \frac{g_2}{g_7}, \quad u_2(\tau) = \frac{g_1}{g_4}.$$

Proof For $\mathbf{a} = (r/15, 0)$ and the variable 15τ we have $q_z = q^r$, so by (8)

$$g_r = -q^{e(r)} (q^r; q^{15})_{\infty} (q^{15-r}; q^{15})_{\infty}, \quad e(r) = \frac{15}{2} B_2\left(\frac{r}{15}\right) = \frac{r^2}{30} - \frac{r}{2} + \frac{5}{4},$$

and $e(2) = \frac{23}{60}$, $e(7) = -\frac{37}{60}$, $e(1) = \frac{47}{60}$, $e(4) = -\frac{13}{60}$. Hence

$$\frac{g_2}{g_7} = q^{23/60+37/60} \frac{(q^2, q^{13}; q^{15})_{\infty}}{(q^7, q^8; q^{15})_{\infty}} = u_1, \quad \frac{g_1}{g_4} = q^{47/60+13/60} \frac{(q, q^{14}; q^{15})_{\infty}}{(q^4, q^{11}; q^{15})_{\infty}} = u_2$$

by (6). $\square$

Lemma 7 (conjugation) Let $\gamma = \begin{pmatrix} 7 & 6 \\ 15 & 13 \end{pmatrix} \in \Gamma_0(15)$. Then for all $\tau \in \mathcal{H}$

$$u_2(\gamma\tau) u_1(\tau) = -1. \quad (10)$$

Proof Put $\tau' = 15\tau$; then $15\gamma\tau = \gamma'\tau'$ with $\gamma' = \begin{pmatrix} 7 & 90 \\ 1 & 13 \end{pmatrix} \in SL_2(\mathbb{Z})$. By Lemma 6 and the first relation in (9),

$$u_2(\gamma\tau) = \frac{g_{(1/15, 0)}(\gamma'\tau')}{g_{(4/15, 0)}(\gamma'\tau')} = \frac{g_{(7/15, 6)}(\tau')}{g_{(28/15, 24)}(\tau')}.$$

Now reduce the indices modulo $\mathbb{Z}^2$ with the second relation in (9): $(7/15, 6) = (7/15, 0) + (0, 6)$ gives the factor $\varepsilon = (-1)^6 e^{\pi i \cdot 6 \cdot 7/15} = e^{4\pi i/5}$, and $(28/15, 24) = (-2/15, 0) + (2, 24)$ gives $\varepsilon = (-1)^{74} e^{-\pi i \cdot 24 \cdot 2/15} = e^{-16\pi i/5} = e^{-6\pi i/5}$. Using $g_{(-2/15, 0)} = -g_{(2/15, 0)}$,

$$u_2(\gamma\tau) = \frac{e^{4\pi i/5} g_7}{-e^{-6\pi i/5} g_2} = -e^{2\pi i} \frac{g_7}{g_2} = -\frac{1}{u_1(\tau)}.$$

(The identity was also checked numerically at six points with 40-digit arithmetic, to within $4 \cdot 10^{-38}$; Section 6.) $\square$

4.2 The zeros

Theorem 8 (i) There is a real number $q_0 \in (-\frac{7}{10}, -\frac{1}{2})$ with $B_2(q_0) = -A_2(q_0)$; numerically $q_0 = -0.570655687750373\dots$. Hence $A_2 + B_2$, $(A_2 + B_2)^3$ and the deferred combination $A_2^3 + B_2^3$ of Entry 8.2.2 all vanish at q_0 , i.e. at $\tau_0 = \frac{1}{2} + iy_0 \in \mathcal{H}$, $y_0 = -\frac{1}{2\pi} \log |q_0| = 0.089281029042\dots$

(ii) Let $\tau_1 = \gamma^{-1}\tau_0 = \frac{13\tau_0 - 6}{-15\tau_0 + 7}$, with γ as in Lemma 7. Then $A_1(\tau_1) = B_1(\tau_1)$. Hence $A_1 - B_1$, $(A_1 - B_1)^3$ and the deferred combination $A_1^3 - B_1^3$ of Entry 8.2.1 all vanish at τ_1 ; numerically $\tau_1 \equiv 0.117021434026 + 0.043690294673i \pmod{1}$, i.e. $q = 0.563611486667 + 0.509757510046i$.

Proof (i) For real $q \in (-1, 0)$ the function u_2 of (6) is a convergent product of positive factors times q , hence continuous and real. Write $u_2(q) = P_K(q) \prod_{k>K} r_k(q)$ with $r_k = (1 - q^{15k+1})(1 - q^{15k+14}) / ((1 - q^{15k+4})(1 - q^{15k+11}))$ and $P_K = q \prod_{k \leq K} r_k$ an exact rational number for rational q . For $|q|^{15(K+1)+1} \leq \frac{1}{2}$ and $k > K$ one has $|\log r_k| \leq 8|q|^{15k+1}$ (each of the four factors $1 - q^m$, $m \geq 15k + 1$, satisfies $|\log(1 - q^m)| \leq 2|q|^m$ because $|q|^m \leq \frac{1}{2}$), so $|\log \prod_{k>K} r_k| \leq S := 8|q|^{15(K+1)+1} / (1 - |q|^{15})$. Exact rational arithmetic gives

$$\begin{aligned} P_4(-\tfrac{1}{2}) &= -0.79954704982494 \dots, & S &< 1.1 \cdot 10^{-22}, \\ P_6(-\tfrac{7}{10}) &= -1.51863908249240 \dots, & S &< 3.1 \cdot 10^{-16}, \end{aligned}$$

so $u_2(-\frac{1}{2}) > -1 > u_2(-\frac{7}{10})$ (the script of Section 6 prints the rigorous enclosures). By the intermediate value theorem there is $q_0 \in (-\frac{7}{10}, -\frac{1}{2})$ with $u_2(q_0) = -1$, i.e. $B_2(q_0) = -A_2(q_0)$, where $A_2(q_0) \neq 0$ by Lemma 5; then $A_2^3 + B_2^3 = A_2^3(1 + u_2^3) = 0$.

(ii) By (10) with $\tau = \tau_1$, $u_1(\tau_1) = -1/u_2(\gamma\tau_1) = -1/u_2(\tau_0) = 1$, i.e. $A_1(\tau_1) = B_1(\tau_1)$, and $A_1(\tau_1) \neq 0$ by Lemma 5. $\square$

Corollary 9 *None of $A_1^3 - B_1^3$, $A_1 - B_1$, $(A_1 - B_1)^3$ (Entry 8.2.1) and $A_2^3 + B_2^3$, $A_2 + B_2$, $(A_2 + B_2)^3$ (Entry 8.2.2) is a theta product (7), for any choice of c, s, m, a_i, b_i, k_i and signs. In particular neither deferred combination has a one-term identity in any basis.*

Proof Each of the six functions vanishes at τ_0 or τ_1 (Theorem 8), whereas a theta product does not vanish anywhere in $\mathcal{H}$ (Lemma 5); two holomorphic functions on $\mathcal{H}$ that agree on a neighbourhood of 0 in the q -plane agree on $\mathcal{H}$. $\square$

Remark 1 (the dichotomy of page 54) In each family Ramanujan wrote as one-term products exactly the combinations that do not vanish — $A_1 + B_1$ (8.2.6), $A_1^3 + B_1^3$ (8.2.8), $A_2 - B_2$ (8.2.10), $A_2^3 - B_2^3$ (8.2.12) — and wrote the vanishing combinations $A_1 - B_1$, $(A_1 - B_1)^3$, $A_2 + B_2$, $(A_2 + B_2)^3$ as sums of two terms, (8.2.7), (8.2.9), (8.2.11), (8.2.13). Corollary 9 shows that this is not a matter of taste: the second group cannot be written with one term. The two deferred combinations belong to the second group, and Theorems 1–2 give their two-term form; what distinguishes them from (8.2.9) and (8.2.13) is that the second term is not a product but $f \cdot a(q^5)$, and Section 5 shows that within the basis of page 54 the only two-term product forms are the trivial ones, $A^3 \mp B^3 = (A^3 \pm B^3) \mp 2B^3$, obtained from the recorded identities (8.2.8) and (8.2.12). The deferral to a note is therefore explained to the extent that a finite search can explain it; a proof that no non-trivial two-term product form exists in general is left open.

Remark 2 The ratios u_1, u_2 are modular units of level 15. A computation like that of Lemma 7 (which we have carried out numerically, to 30 digits) shows how $\Gamma_0(15)$ acts on them according to the residue of the upper-left entry modulo 15: for ± 1 both are invariant, e.g. under $(\frac{1}{15} \frac{1}{16})$; for ± 4 each is inverted, e.g. $u_1(\gamma\tau) = 1/u_1(\tau)$ for $\gamma = (\frac{4}{15} \frac{1}{4})$; for ± 2 and ± 7 the two are interchanged up to sign or inversion, e.g. $u_2(\gamma\tau) = -u_1(\tau)$ for $\gamma = (\frac{2}{15} \frac{1}{8})$ and $u_2(\gamma\tau) = -1/u_1(\tau)$ for the γ of Lemma 7. The mirror symmetry $f(-q^3, -q^{12}) \leftrightarrow f(-q^6, -q^9)$, $f(-q, -q^4) \leftrightarrow f(-q^2, -q^3)$ between the two entries, under which (8.2.8) corresponds to (8.2.12) and (8.2.9) to (8.2.13), is the shadow of this conjugation; in particular the two deferrals are two instances of one phenomenon, and both lost notes are the same circular summation (1) with the same modulus $ab = q^5$.

5 A finite search for two-term product forms

Corollary 9 rules out one-term forms completely. For two-term forms, which the recorded identities (8.2.9) and (8.2.13) do possess, we report a finite search.

5.1 Non-periodicity of the quadratic cofactors

Write $A^3 + B^3 = (A + B)(A^2 - AB + B^2)$ and $A^3 - B^3 = (A - B)(A^2 + AB + B^2)$. A power series with integer coefficients and constant term 1 has a unique representation $\prod_{n \geq 1} (1 - q^n)^{-c_n}$. If (c_n) is periodic with period M , the series equals $\prod_{a=1}^M (q^a; q^M)_\infty^{-c_a}$, a generalized eta-quotient; when $c_a = c_{M-a}$ for $1 \leq a < M$ it is a theta product in the sense of (7), since $(q^a; q^M)_\infty (q^{M-a}; q^M)_\infty = f(-q^a, -q^{M-a}) / (q^M; q^M)_\infty$. Conversely, a theta product (7) with integer a_i, b_i and $c = 1, s = 0$ is, by the Jacobi triple product and $1 + x = (1 - x^2)/(1 - x)$, a finite product of factors $(q^a; q^M)_\infty^{\pm 1}$, so its (c_n) is periodic. Computing (c_n) for $n \leq 380$ by the logarithmic-derivative recursion we find, in each entry, a sharp split:

Entry	combination	(c_n)
8.2.1	$A^2 - AB + B^2$ (factor of the recorded (8.2.8))	periodic mod 15
8.2.1	$A^2 + AB + B^2$ (factor of the lost $A^3 - B^3$)	not periodic
8.2.2	$A^2 + AB + B^2$ (factor of the recorded (8.2.12))	periodic mod 15
8.2.2	$A^2 - AB + B^2$ (factor of the lost $A^3 + B^3$)	not periodic

(“Not periodic” means: no period ≤ 60 holds for $n \leq 380$; the periodic cases were checked from $n = 1$.) The non-periodic cofactors are $A_1^2(1 + u_1 + u_1^2)$ and $A_2^2(1 - u_2 + u_2^2)$, which vanish where $u_1 \in \{\omega, \omega^2\}$, resp. $u_2 \in \{-\omega, -\omega^2\}$, $\omega = e^{2\pi i/3}$; numerically such points exist (e.g. $u_1(\tau) = \omega$ at $\tau \approx 0.29434 + 0.06065i$), which is consistent with the table, but we have not proved it, and the table rules out any identity of the shape $A^3 \mp B^3 = (\text{linear factor known on p. 54}) \times (\text{theta product})$ only within the computed range.

5.2 Systematic search over theta-monomial bases

We searched for representations

$$A^3 - B^3 = \sum_{i=1}^r c_i q^{s_i} M_i, \quad r \leq 2, \quad c_i \in \{\pm 1, \pm 2, \pm 3\}, \quad s_i \in \{0, \dots, 3\},$$

where each M_i is a monomial of total degree 3 with exponents in $\{-1, 1, 2, 3\}$ in at most three functions drawn from the 12-function basis

$$\begin{aligned} & f(-q, -q^4), f(-q^2, -q^3), f(-q^3, -q^{12}), f(-q^6, -q^9), f(-q^5), \\ & f(-q, -q^{14}), f(-q^2, -q^{13}), f(-q^4, -q^{11}), f(-q^7, -q^8), f(-q), f(-q^3), f(-q^{15}) \end{aligned}$$

— every theta function occurring in Entries 8.2.1–8.2.2 together with the relevant η -type functions (2344 monomials; hash-matching on the first 80 coefficients, surviving candidates confirmed on 260). The search was repeated in the $q^{1/3}$ -lattice (setting $q \mapsto q^3$ and adjoining the functions $f(-q^{a/3}, -q^{b/3})$ occurring in (8.2.7) and (8.2.11): 20 basis functions, 11800 monomials, shifts $s_i \in \{0, \dots, 9\}$), and with the three-term shape $c_1 q^{s_1} M_1 + c_2 q^{s_2} M_2 \pm 3q^s AB M_3$ suggested by Theorems 3–4.

Control of the instrument. Run on the recorded combinations, the same search re-finds Ramanujan’s identities exactly: for $(A - B)^3$ of Entry 8.2.1 it returns the two-term representation (8.2.9), and in the $q^{1/3}$ -lattice it returns (8.2.7) for $A - B$.

Result. For $A^3 - B^3$ (Entry 8.2.1) and $A^3 + B^3$ (Entry 8.2.2) the search in the integer basis returns exactly three representations each, all trivial: the definition itself, $f^3(-q^7, -q^8) - q^3 f^3(-q^2, -q^{13})$, resp. $f^3(-q^4, -q^{11}) + q^3 f^3(-q, -q^{14})$, and, once the coefficients ± 2 are admitted, the two rewritings $A^3 \mp B^3 = (A^3 \pm B^3) \mp 2B^3 = 2A^3 - (A^3 \pm B^3)$ of the recorded one-term identities (8.2.8) and (8.2.12); each candidate was confirmed on all 260 coefficients. In the $q^{1/3}$ -lattice (trivial forms excluded) and with the three-term shape the search returns nothing. This is consistent with Theorems 1–2: the missing identities need the binary theta series $a(q^5)$, which lies outside the product basis. We state the scope honestly: this is a finite search over an explicit basis and coefficient range; the one-term case is settled by Corollary 9, the non-trivial two-term case in general is not.

6 The verification scripts

All twelve identities of Sections 2–3 — (8.2.6)–(8.2.13), (4)–(5) and (2)–(3) — are certified by one self-contained Python script `verifier.py` (152 lines, no dependencies). It works in the variable $y = q^{1/3}$ with exact integer arithmetic; every identity involving a quotient is verified after cross-multiplication, so no division of power series occurs. Each identity is checked to 400 coefficients of q by default, and the run reported here used 2000 coefficients (6003 powers of y ; total runtime about one second). The script prints one MOS/FARQ (match/mismatch) line per identity and exits nonzero on any mismatch; the archived run prints 12/12 MOS. A second, independent script `sonli_tekshir.py` evaluates all identities as printed (with division and $q^{1/3}$) at five values of q with 50-digit arithmetic; the largest relative discrepancy is $2 \cdot 10^{-49}$. A third script `ildiz.py` checks the foundations independently: it compares every theta series used with its Jacobi triple product (3000 coefficients), verifies the circular summation (1) itself as an identity of two-variable integer power series in independent variables a, b (all monomials $a^i b^j$ with $i + j \leq 150$, where the brace is replaced by $a(ab)$), and checks $a^3 = b^3 + c^3$ in the division-free form $a(q)^3(q; q)_\infty^3(q^3; q^3)_\infty^3 = (q; q)_\infty^{12} + 27q(q^3; q^3)_\infty^{12}$ (3000 coefficients). `ildiz_gp.py` re-implements the theta series, Theorems 1–2, both by-products and the Borwein identity from scratch in PARI/GP (1500 coefficients), independently of the Python code.

The script `nol_sertifikat.py` accompanies Section 4. Its Part A is the proof of Theorem 8(i): it computes $P_4(-\frac{1}{2})$ and $P_6(-\frac{7}{10})$ as exact rational numbers (denominators of 97 and 623 digits), bounds the tails by the inequality stated in the proof, and prints rigorous enclosures of $u_2(-\frac{1}{2})$ and $u_2(-\frac{7}{10})$ on the two sides of -1 . Its

Part B is a check, at 40 digits, of Lemma 7 at six points, of the location of τ_0 and τ_1 , of $|A_1(\tau_1) - B_1(\tau_1)| < 10^{-30}$, and of the non-vanishing of the recorded products (8.2.6), (8.2.8), (8.2.10), (8.2.12) at the same points.

Use of AI tools. In accordance with the journal’s policy on large language models, we state that generative AI (Claude, Anthropic) was used to write and run the verification and search programs described in this section and in Sections 4–5, to carry out the numerical exploration that located the zeros of Section 4, and to help prepare the text. The author directed the work, checked every statement against the cited sources and the program output, and takes full responsibility for the content of the paper.

Declarations

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Data availability. No data were generated beyond the program outputs described in Section 6.

Materials availability. Not applicable.

Code availability. The verification scripts and the search scripts of Section 5 are archived at Zenodo: <https://doi.org/10.5281/zenodo.23030496>.

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References

- [1] S. Ramanujan, *The Lost Notebook and Other Unpublished Papers*, Narosa, New Delhi, 1988.
- [2] G. E. Andrews and B. C. Berndt, *Ramanujan’s Lost Notebook, Part II*, Springer, New York, 2009 (Chapter 8).
- [3] B. C. Berndt, *Ramanujan’s Notebooks, Part III*, Springer, New York, 1991 (Chapter 20, Entries 10(i)–(iv)).
- [4] S. H. Son, *Circular summations of theta functions in Ramanujan’s lost notebook*, Ramanujan J. **8** (2004), 235–272, doi:10.1023/B:RAMA.0000040483.55191.d1.
- [5] G. E. Andrews and B. C. Berndt, *Ramanujan’s Lost Notebook, Part III*, Springer, New York, 2012.
- [6] G. E. Andrews and B. C. Berndt, *Ramanujan’s Lost Notebook, Part IV*, Springer, New York, 2013.

- [7] J. M. Borwein and P. B. Borwein, *A cubic counterpart of Jacobi's identity and the AGM*, Trans. Amer. Math. Soc. **323** (1991), 691–701, doi:10.1090/S0002-9947-1991-1010408-0.
- [8] J. M. Borwein, P. B. Borwein and F. G. Garvan, *Some cubic modular identities of Ramanujan*, Trans. Amer. Math. Soc. **343** (1994), 35–47, doi:10.1090/S0002-9947-1994-1243610-6.
- [9] H. H. Chan, Z.-G. Liu and S. T. Ng, *Circular summation of theta functions in Ramanujan's lost notebook*, J. Math. Anal. Appl. **316** (2006), 628–641, doi:10.1016/j.jmaa.2005.05.015.
- [10] J.-K. Ge and Q.-M. Luo, *Some extensions for Ramanujan's circular summation formula*, arXiv:1901.09820 (2019).
- [11] D. S. Kubert and S. Lang, *Modular Units*, Grundlehren der mathematischen Wissenschaften **244**, Springer, New York, 1981.
- [12] S. H. Son, *Cubic identities of theta functions*, Ramanujan J. **2** (1998), 303–316, doi:10.1023/A:1009751614537.